

Summarization

Md. Shad Akhtar
Assistant Professor, IIIT Delhi

<http://faculty.iiitd.ac.in/~shad.akhtar/>
<http://lcs2.iiitd.edu.in/>

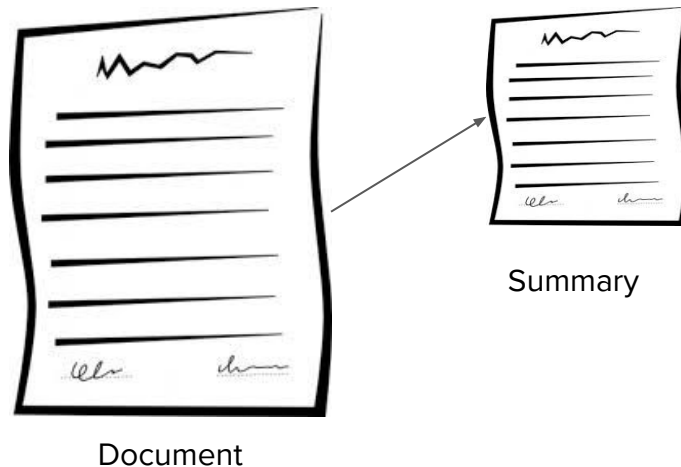


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Summarization

- Task: produce an abridged version of a text while retaining the key and relevant information
- Applications:
 - outlines or abstracts of any document, article, etc
 - summaries of chat and email
 - action items from a meeting
 - simplifying text by compressing sentences
 - and many more....



Multiple Perspectives

- Input:
 - Single document summarization (SDS)
 - Multiple-document summarization (MDS)
- Output:
 - Extractive
 - Abstractive
- Focus:
 - Generic (unconditioned)
 - Query-focused (conditioned)
- Evaluation factor

Perspective: Input

Document

Cambodian leader Hun Sen on Friday rejected opposition parties' demands for talks outside the country, accusing them of trying to "internationalize" the political crisis.

Government and opposition parties have asked King Norodom Sihanouk to host a summit meeting after a series of post-election negotiations between the two opposition groups and Hun Sen's party to form a new government failed.

Opposition leaders Prince Norodom Ranariddh and Sam Rainsy, citing Hun Sen's threats to arrest opposition figures after two alleged attempts on his life, said they could not negotiate freely in Cambodia and called for talks at Sihanouk's residence in Beijing. Hun Sen, however, rejected that.

I would like to make it clear that all meetings related to Cambodian affairs must be conducted in the Kingdom of Cambodia," Hun Sen told reporters after a Cabinet meeting on Friday. "No-one should internationalize Cambodian affairs.

It is detrimental to the sovereignty of Cambodia," he said. Hun Sen's Cambodian People's Party won 64 of the 122 parliamentary seats in July's elections, short of the two-thirds majority needed to form a government on its own. Ranariddh and Sam Rainsy have charged that Hun Sen's victory in the elections was achieved through widespread fraud. They have demanded a thorough investigation into their election complaints as a precondition for their cooperation in getting the national assembly moving and a new government formed

Summary

Cambodian government rejects
opposition's call for talks abroad

Documents

Fingerprints and photos of two men who boarded the doomed Malaysia Airlines passenger jet are being sent to U.S. authorities so they can be compared against records of known terrorists and criminals. The cause of the plane's disappearance has baffled investigators and they have not said that they believed that terrorism was involved, but they are also not ruling anything out. The investigation into the disappearance of the jetliner with 239 passengers and crew has centered so far around the fact that two passengers used passports stolen in Thailand from an Austrian and an Italian. The plane which left Kuala Lumpur, Malaysia, was headed for Beijing. Three of the passengers, one adult and two children, were American.

(CNN) -- A delegation of painters and calligraphers, a group of Buddhists returning from a religious gathering in Kuala Lumpur, a three-generation family, nine senior travelers and five toddlers. Most of the 227 passengers on board missing Malaysia Airlines Flight 370 were Chinese, according to the airline's flight manifest. The 12 missing crew members on the flight that disappeared early Saturday were Malaysian. The airline's list showed the passengers hailed from 14 countries, but later it was learned that two people named on the manifest -- an Austrian and an Italian -- whose passports had been stolen were not aboard the plane. The plane was carrying five children under 5 years old, the airline said.

:

Vietnamese aircraft spotted what they suspected was one of the doors belonging to the ill-fated Malaysia Airlines Flight MH370 on Sunday, as troubling questions emerged about how two passengers managed to board the Boeing 777 using stolen passports. The discovery comes as officials consider the possibility that the plane disintegrated mid-flight, a senior source told Reuters. The state-run Thanh Nien newspaper cited Lt. Gen. Vo Van Tuan, deputy chief of staff of Vietnam's army, as saying searchers in a low-flying plane had spotted an object suspected of being a door from the missing jet. It was found in waters about 56 miles south of Tho Chu island, in the same area where oil slicks were spotted Saturday.

Summary

Flight MH370, carrying 239 people vanished over the South China Sea in less than an hour after taking off from Kuala Lumpur, with two passengers boarded the Boeing 777 using stolen passports. Possible reasons could be an abrupt breakup of the plane or an act of terrorism. The government was determining the "true identities" of the passengers who used the stolen passports. Investigators were trying to determine the path of the plane by analysing civilian and military radar data while ships and aircraft from seven countries scouring the seas around Malaysia and south of Vietnam.

Single document summarization (SDS)

Multiple-document summarization (MDS)

Another dimension can be multilingual and multi-document summarization.

Perspective: Output

Pros and cons of these two approaches??

- Extractive summarization
 - Select from the source text spans that capture the key information
- Abstractive summarization
 - Generate new text that encapsulates the key information from the source text

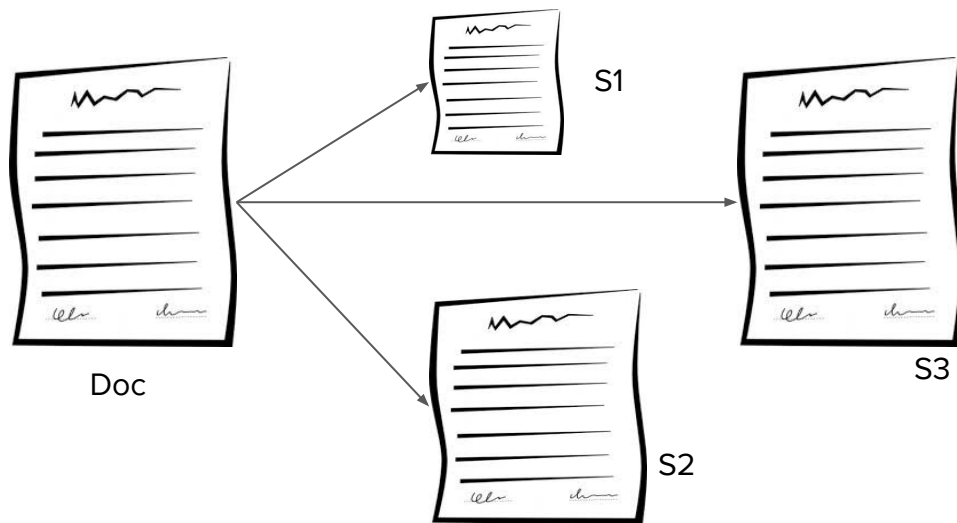


Perspective: Focus

- Generic
 - Summarize the content of the doc(s)
- Query-focused
 - Summarize a doc w.r.t. a user's query.
 - There may be other critical information but those are not required to serve the given query.
 - Also known as controlled summarization
 - In a way, we are controlling what we need as a summary
- Peer-review of project report:
 - a. **Description:** Briefly describe the report in your own words. You should not copy the abstract. (Generic)
 - b. **Strength of the paper (at least 2):** What do you think about the strength of the paper? (Query-focused / Controlled)
 - c. **Weakness of the paper (at least 2):** What do you think is missing and/or has some unintuitive/illogical components (approach, result, architecture, etc.)? Suggestions, if any. (Query-focused / Controlled)

Perspective: Evaluation Factor

- How much summary is a good summary?
 - Difficult to say (depends on the requirement); however, typically, the relevant/desired content should be captured.
- If a document is 1000 word long and it has a few (let's say 10) relevant information, then what should be the size of summary?



Assuming all three summaries captured the relevant information, which one is better?

It depends on the *Budget*.

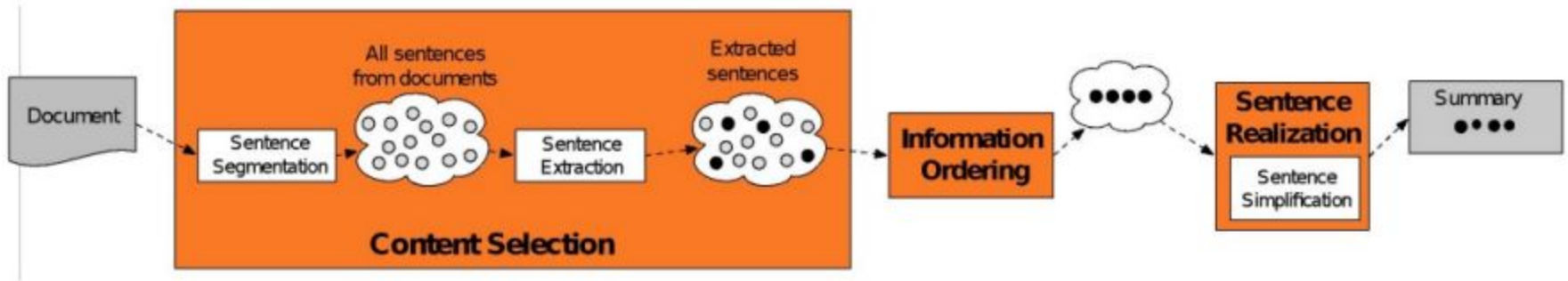
Summary evaluation

- Extractive
 - How many sentences are overlapping with the reference summary?
 - F1-score, Accuracy, etc.
- Abstractive
 - How good the generated summary is against the reference summary?
 - Lexical:
 - BLEU, ROUGE, METEOR
 - Sematic:
 - BERTScore and others.
- For conditional summarization, you should define/use a metric which validates the relevancy of conditions captured in the summary.
 - E.g.,
 - In peer-review setting, is there any positive statement, acknowledgement, or praising in the strength of the paper summary?

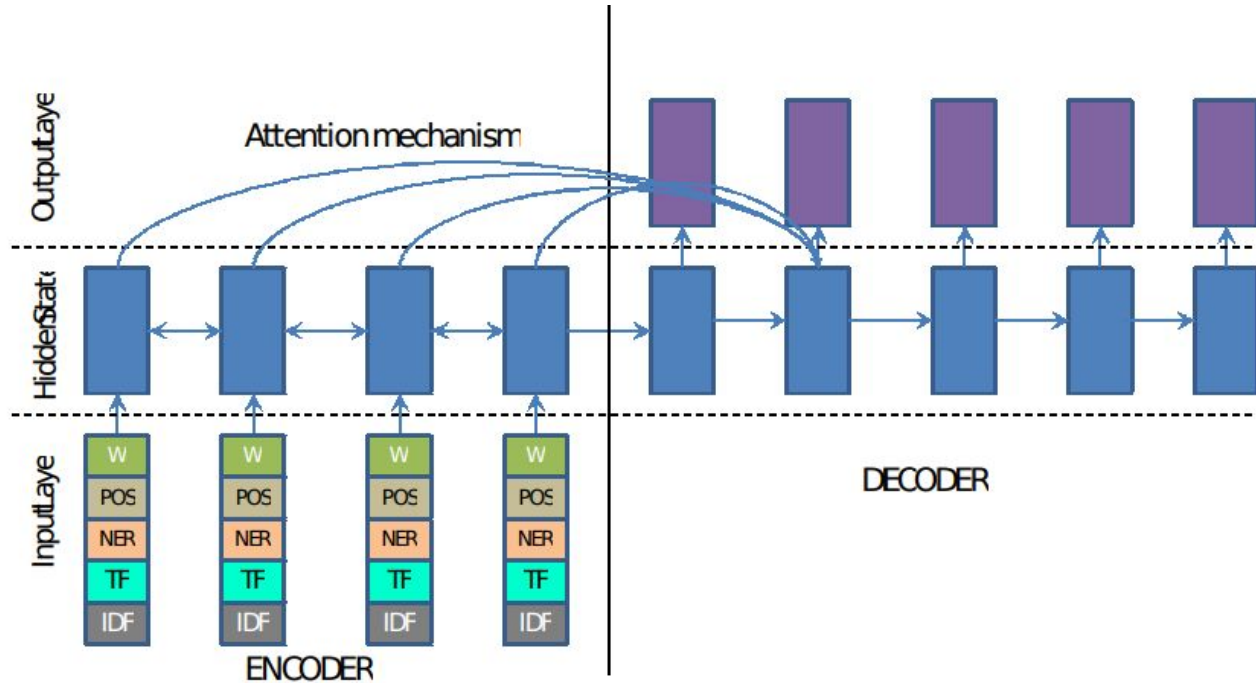
Approaches

Extractive Methods

- Content Selection:
- Information ordering
- Sentence realization (Optional)

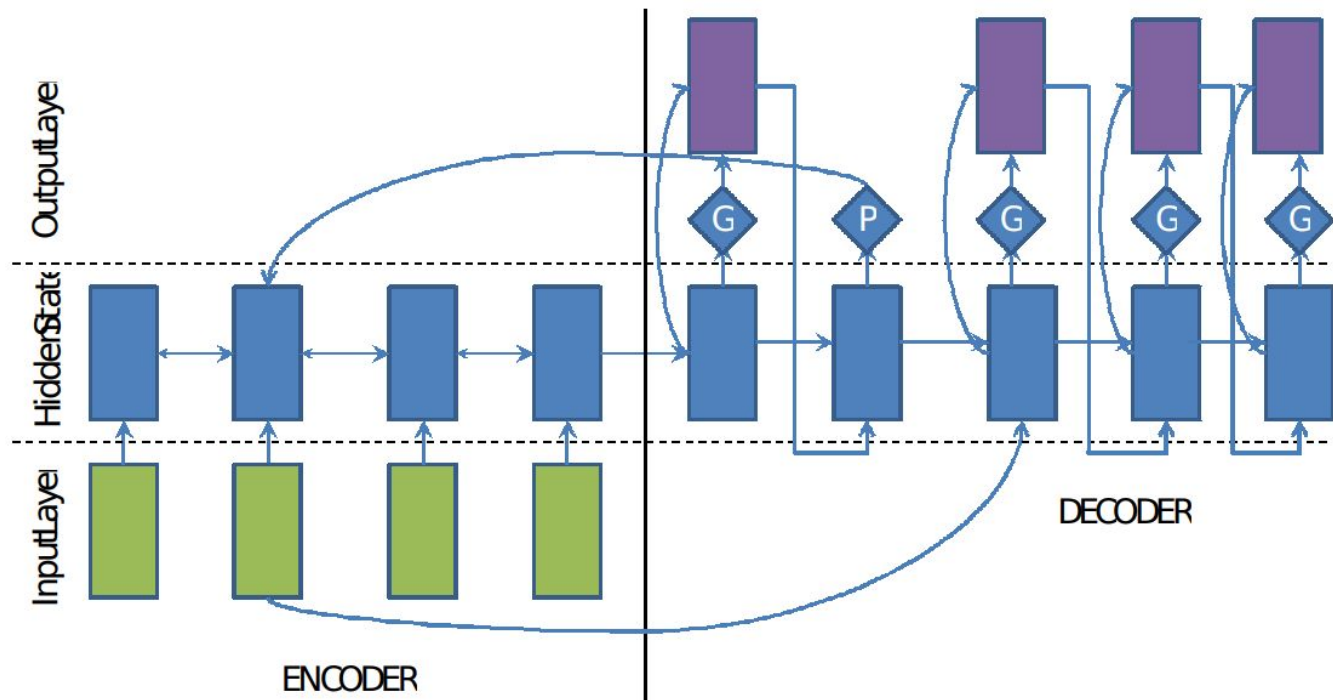


Abstractive Text Summarization using Sequence-to-sequence RNNs and Beyond [Nallapati et al., CoNLL-2016]



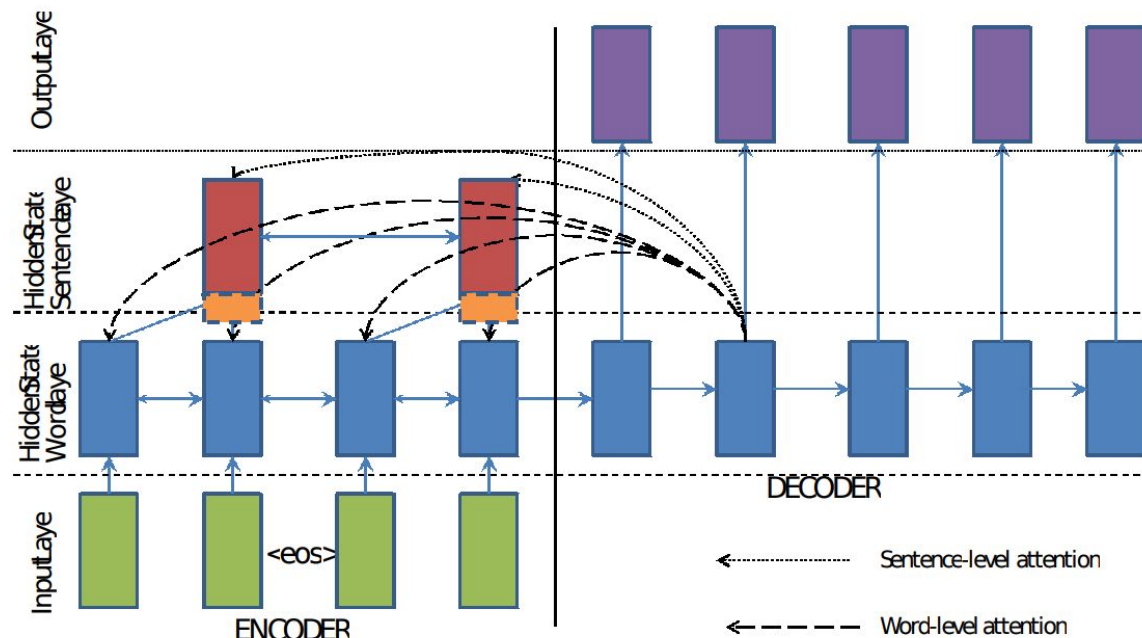
Typical Encoder-Attend-Decoder architecture.

Abstractive Text Summarization using Sequence-to-sequence RNNs and Beyond [Nallapati et al., CoNLL-2016]



An intuitive way to handle OOV words is to simply point to their location in the source document. G: Typical generation using softmax over vocabulary; P: Rare words, point to the word in the document.

Abstractive Text Summarization using Sequence-to-sequence RNNs and Beyond [Nallapati et al., CoNLL-2016]



Hierarchical encoder with hierarchical attention

Get To The Point: Summarization with Pointer-Generator Networks (PGN) [See et al., ACL-2017]

- Aims to handle the factual information through a mixture approach of copying/pointing and generating.

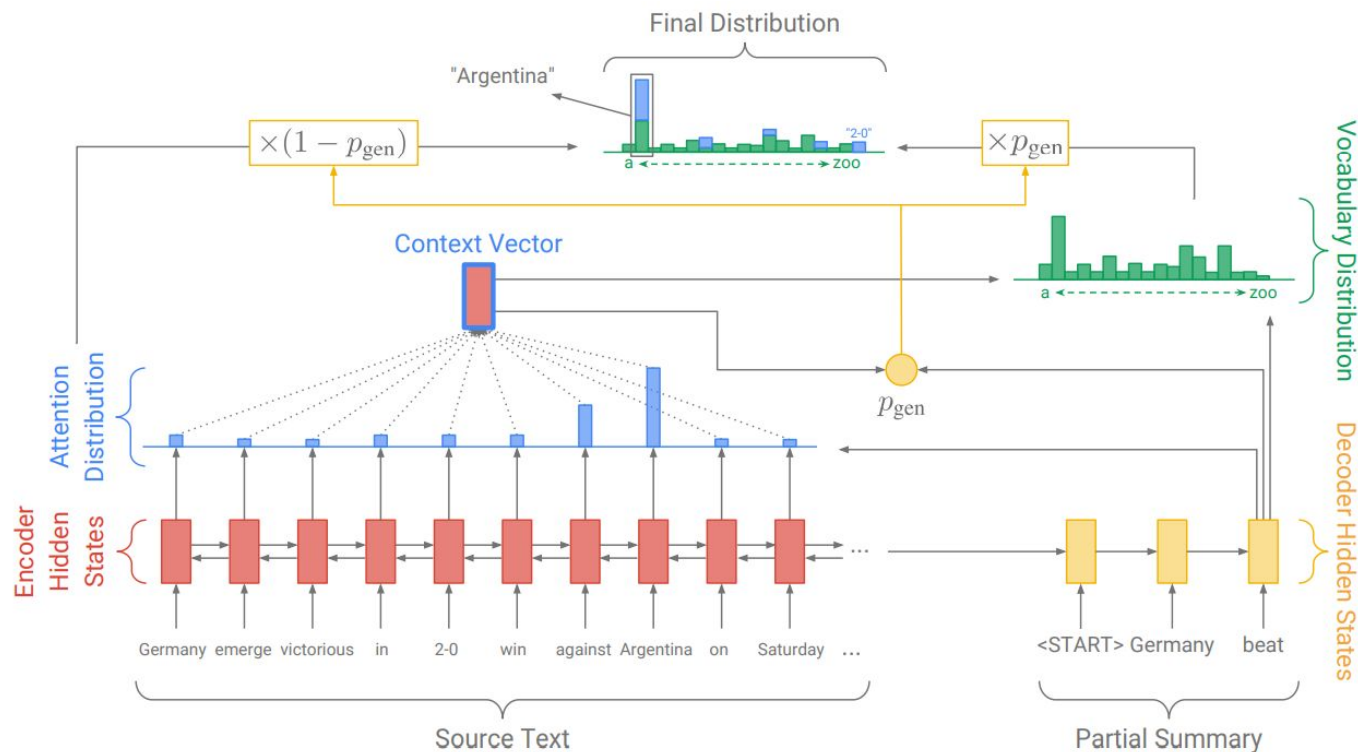
Article: andy murray (...) is into the semi-finals of the miami open , but not before getting a scare from 21 year-old austrian dominic thiem, who pushed him to 4-4 in the second set before going down 3-6 6-4, 6-1 in an hour and three quarters. (...)

Summary: andy murray **defeated** dominic thiem 3-6 6-4, 6-1 in an hour and three quarters.

Article: (...) wayne rooney smashes home during manchester united 's 3-1 win over aston villa on saturday. (...)

Summary: manchester united **beat** aston villa 3-1 at old trafford on saturday.

Get To The Point: Summarization with Pointer-Generator Networks (PGN) [See et al., ACL-2017]

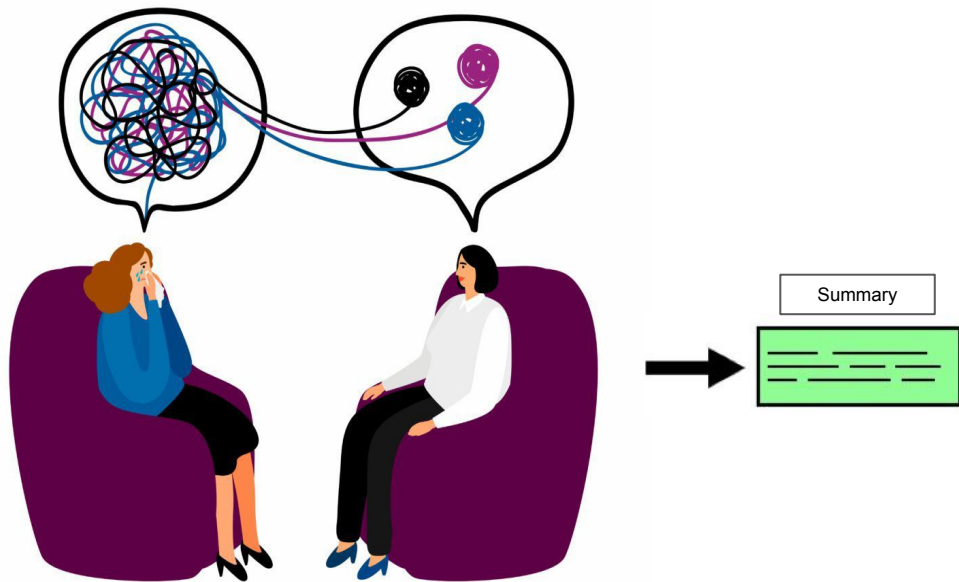


P_{gen} is used as a soft switch to choose between generating a word from the vocabulary by sampling from P_{vocab} , or copying a word from the input sequence by sampling from the attention distribution.

Controlled Summary Generation

Counseling Summarization using Mental Health Knowledge

Guided Utterance Filtering [Srivastava et al., KDD-2022]



- **Understanding Client**

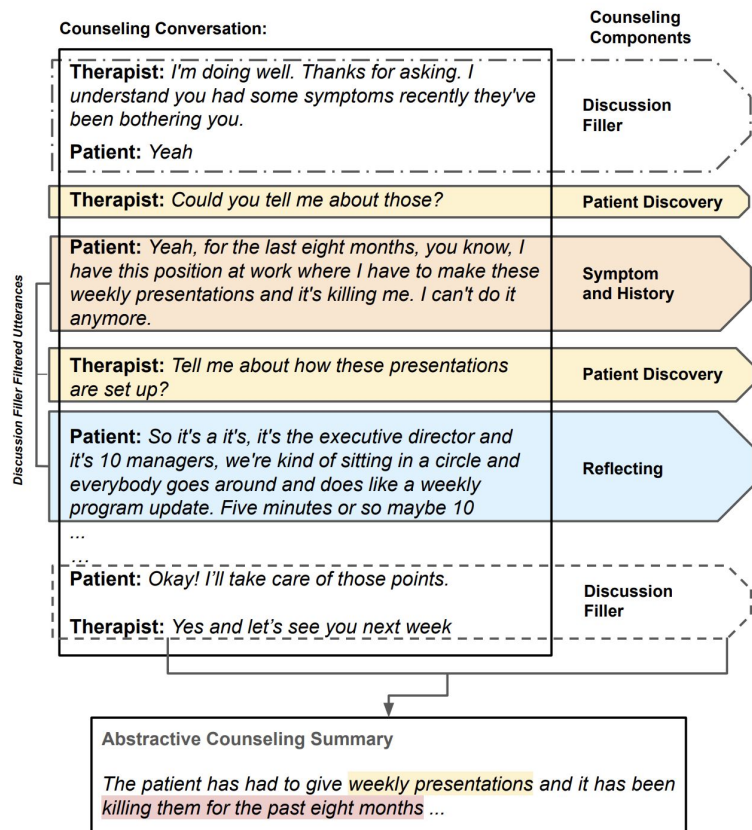
- Topic
- Symptoms / Reasoning
- Story
- Routine Conversation

- **Counseling Summary**

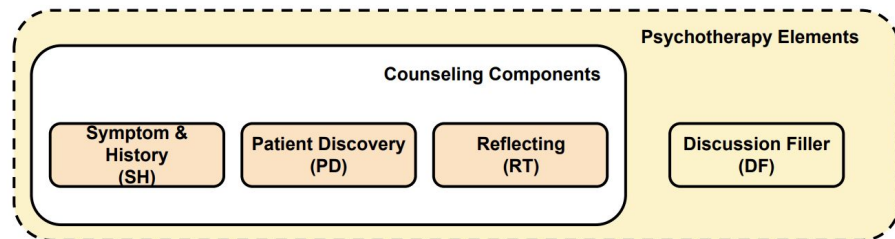
Use of state-of-the-art deep learning models to generate quality summary.

Counseling Summarization using Mental Health Knowledge

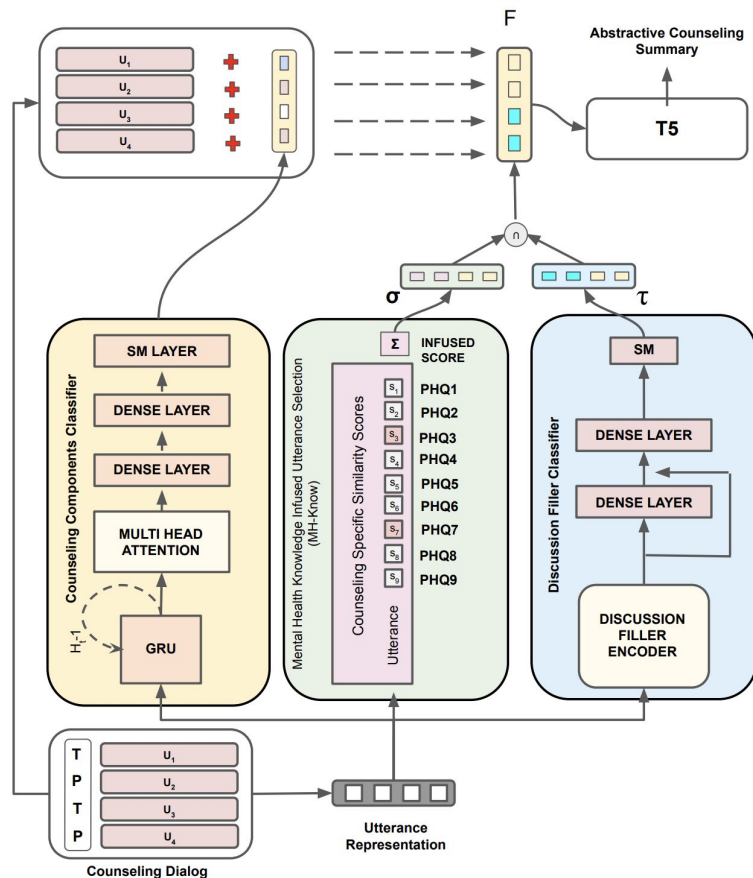
Guided Utterance Filtering [Srivastava et al., KDD-2022]



- HOPE - Cognitive behaviour therapy
- A session contains psychotherapy elements viz. symptoms, history of mental health issues, or the discovery of the patient's behavior along with some discussion filler.
- We refer to these important components as **counseling components**.



Counseling Summarization using Mental Health Knowledge Guided Utterance Filtering [Srivastava et al., KDD-2022]



Complete architecture is divided into three filtering modules:

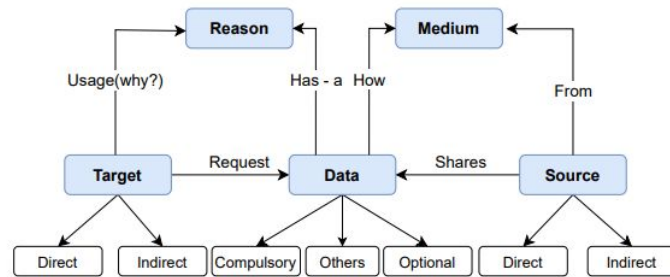
- *Discussion Filler Classifier (DFC)*
- *Mental Health Knowledge Infused Utterance Selection (MHKnow)*
- *Counseling Components Classifier (CCC)*

EROS: Entity-Driven Controlled Policy Document Summarization

[Singh et al., LREC-COLING-2024]

Entity	Explanation
Data Compulsory	Data which is compulsory for the source to enter
Data Optional	Data in which the source has the option to provide to the target
Data Others	Information that belongs to the source
Source Direct	The entity that directly provides the data to the target
Source Indirect	The entity that indirectly provides the data to the target
Target Direct	The entity to which the source directly provides the data
Target Indirect	The entity to which the source indirectly provides the data
Reason	Reason for why the data is being collected by the target
Medium	How the data will be collected by the target

(a) Entity labels with their explanation

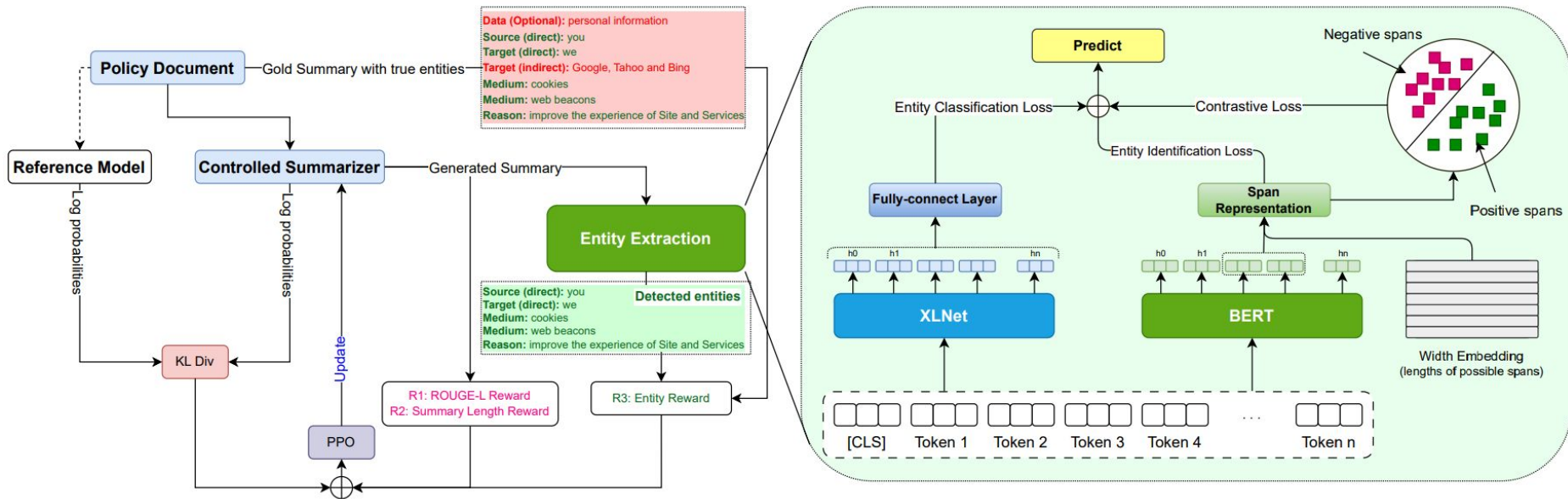


(b) Relation between entities

Paragraph: When {you}_{source-direct} {visit the site}_{medium}, {we}_{target-direct} also collect {web site usage information , the type and version of browser and operating system}_{data-compulsory} {you}_{source-direct} use, {if you arrived at trainchinese.com via a link from another website , the URL of the linking page}_{data-compulsory}. {We}_{target-direct} use this information to {ensure our site is compatible with the browsers used by most of our visitors}_{reason} and to {improve the customer experience}_{reason}.

EROS: Entity-Driven Controlled Policy Document Summarization

[Singh et al., LREC-COLING-2024]



EROS: Entity-Driven Controlled Policy Document Summarization

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Ref Summary: When you register with the Site or use any of our Services, you may be asked to provide us with Personal Information. It is entirely optional to provide this information. If you do not provide the requested information, you may be unable to use some or all of the Site's features. To improve the Site and Services, we use cookies (a small text file placed on your computer to identify your computer and browser) and Web beacons (electronic files placed on a Web site that monitor usage). If you delete or disable cookies, some features of the Site or Services may not function properly. You have the option of making some of your Personal Information available to others. Users of the Site and commercial search engines such as Google, Yahoo!, and Bing may access the information you choose to make available. Some third-party services may provide us with information from your accounts, allowing us to improve and personalise your experience on the Site. When you visit our Site, we may allow third-party companies to serve ads and/or collect certain anonymous information....

Entities in Ref summary: **Data compulsory:** [e-mail, personally identifiable information]; **Data others:** [personal information]; **Source Direct:** [you, info@day-finder.com]; **Target Direct:** [we, us]; **Target Indirect:** [service providers, third parties]; **Medium:** [register with the Site or use any of our Services, cookies, web beacons, visit our site]; **Reason:** [improve the site and services, improve and personalise your experience on the Site, Usage Data regarding the Site and Services].

Entities in BART summary: **Data compulsory:** [None]; **Data others:** [None]; **Source Direct:** [you, info@day-finder.com]; **Target Direct:** [we, us]; **Target Indirect:** [None]; **Medium:** [using the site or services, cookies, web beacons, visit our site]; **Reason:** [improve the experience of the site and services].

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Entities in EROS w/o R3 summary: **Data compulsory:** [e-mail, personally identifiable information]; **Data others:** [personal information]; **Source Direct:** [you]; **Target Direct:** [we, us]; **Target Indirect:** [service providers]; **Medium:** [register with the Site or use any of our Services, cookies, web beacons, visit our site]; **Reason:** [None].

Entities in EROS summary: **Data compulsory:** [e-mail, personally identifiable information]; **Data others:** [personal information]; **Source Direct:** [you, info@day-finder.com]; **Target Direct:** [we, us]; **Target Indirect:** [service providers, third parties]; **Medium:** [register with the Site or use any of our Services, cookies, web beacons, visit our site]; **Reason:** [Usage Data regarding the Site and Services].

Table : Qualitative analysis of generated summaries. Due to space constraints, we could not provide the generated summaries of models. **source direct**, **target direct**, **medium - blue**, **reason**, **data**, **data compulsory**, **target indirect**, **data optional**. Best viewed in color.

Thank You!

